

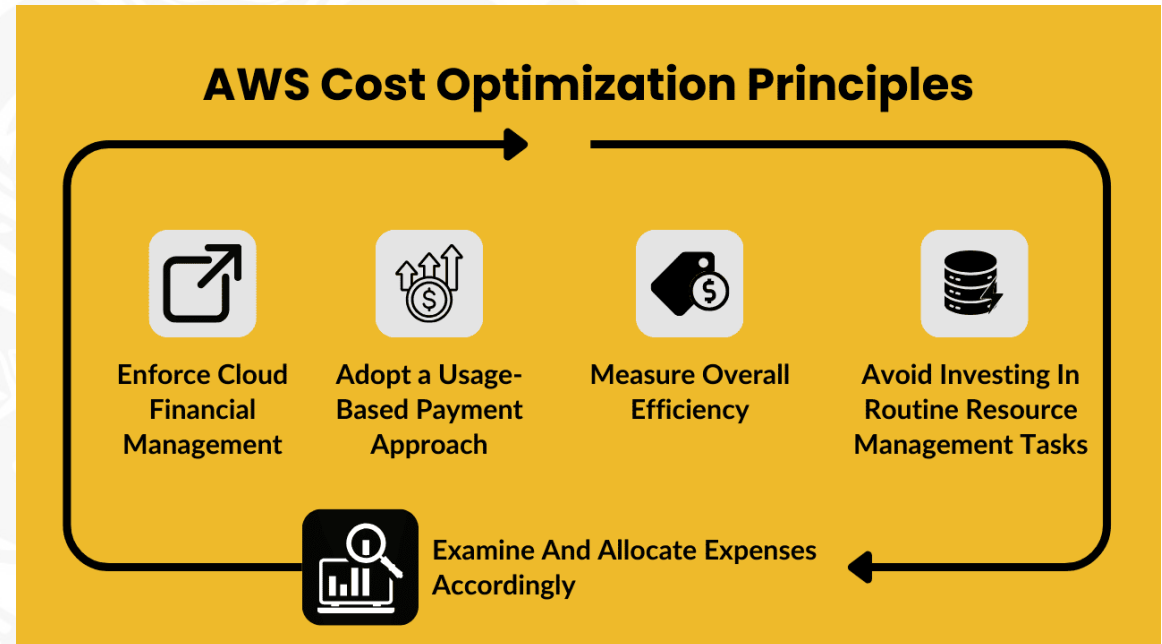


CyberDojo : The School of
Cyberdefense

SAA-C03
Module 4:
Design Cost
Optimized
Architectures

Module 4: Design Cost-Optimized Architectures

- Section 4.1: Cost-Optimized Storage (S3 storage classes, lifecycle policies, EBS optimization)
- Section 4.2: Cost-Optimized Compute (Pricing models, Savings Plans, Spot Instances, Compute Optimizer)
- Section 4.3: Cost-Optimized Database (RDS pricing, Aurora Serverless, DynamoDB capacity modes)
- Section 4.4: Cost-Optimized Network (Data transfer costs, VPC endpoints, NAT Gateway vs. NAT Instance)



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S3 Storage Classes Overview - Complete Comparison

Storage Class	Durability	Availability	Min Duration	Retrieval	Use Case
S3 Standard	11 9's	99.99%	None	ms	Frequently accessed data
S3 Intelligent-Tiering	11 9's	99.9%	30 days	ms	Unknown/mixed access patterns
S3 Standard-IA	11 9's	99.9%	30 days	ms	Infrequent access (backups)
S3 One Zone-IA	11 9's	99.5%	30 days	ms	Recreatable infrequent data
S3 Glacier Instant	11 9's	99.9%	90 days	ms	Archive, immediate access
S3 Glacier Flexible	11 9's	99.99%	90 days	min-hours	Archive, slower retrieval OK
S3 Glacier Deep Archive	11 9's	99.99%	180 days	12+ hours	Long-term cold storage



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S3 Storage Class Selection - Decision Framework

Question	Answer	Recommended Class
How often is data accessed?	Frequently (daily/hourly)	S3 Standard
	Infrequently (monthly)	S3 Standard-IA
	Rarely (yearly)	Glacier Flexible/Deep
How fast is retrieval needed?	Milliseconds	Standard, IA, Glacier Instant
	Minutes to hours	Glacier Flexible
	12+ hours	Glacier Deep Archive
Can data be recreated?	Yes	One Zone-IA (cheaper)
	No	Standard, IA, Glacier (Multi-AZ)
Unknown access pattern?	Mixed/unknown	Intelligent-Tiering

SAA Exam Focus:

- **Durability = 11 9's for ALL classes** (exception: One Zone-IA = same durability but only 1 AZ)
- **Availability** varies (99.99% Standard vs. 99.5% One Zone-IA)
- **Minimum duration** = Early deletion fee if deleted before (30, 90, 180 days)



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S3 Lifecycle Policies - Transition and Expiration

What are Lifecycle Policies? Rules to automatically transition or expire objects

Two Types of Actions:

Action	What it does	Example
Transition	Move to cheaper storage class	Standard → IA after 30 days
Expiration	Delete objects	Delete after 365 days

- **SAA Exam Focus:**

- Transition = Move (cost optimization)
- Expiration = Delete (cleanup)
- Cannot transition from Glacier to Standard (one-way trip!)
- Minimum days between transitions = 30 days (can't transition daily)

Common Lifecycle Pattern (Exam Favorite):

Day 0-30:	S3 Standard (hot)
Day 31-90:	S3 Standard-IA (warm)
Day 91-365:	S3 Glacier Flexible (cold)
Day 366+:	Expire (delete)



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EBS Cost Optimization - gp3 as Default

EBS Volume Type Cost Comparison:

Volume Type	Cost (us-east-1)	IOPS Baseline	Throughput Baseline
gp3	\$0.08/GB-month	3,000	125 MB/s
gp2	\$0.10/GB-month	3 IOPS/GB	250 MB/s (at 1,000+ IOPS)
io1 (Provisioned)	\$0.125/GB-month + \$0.065/provisioned IOPS	Configurable	Configurable

EBS Cost Optimization Tips (Exam Favorites):

Tip	Savings
Use gp3 instead of gp2	20% cheaper (same or better performance)
Delete unused volumes	100% savings (stop paying for idle)
Snapshots for backup, not clones	Incremental snapshots cheaper than duplicate volumes
EBS lifecycle policies	Automatically delete old snapshots
Right-size volumes	Don't provision more than needed

- **Delete unused EBS volumes** (common cost optimization question)
- **Snapshots are incremental** (only changed blocks)



EFS Cost Optimization - IA Tier and Lifecycle Management

EFS Storage Classes:

Class	Use Case	Cost (us-east-1)
EFS Standard	Frequently accessed data	\$0.30/GB-month
EFS Infrequent Access (IA)	Less frequently accessed	\$0.025/GB-month (88% cheaper!)
EFS Archive	Long-term cold storage	\$0.008/GB-month

EFS Cost Optimization Example:

100 GB of files accessed daily → EFS Standard: \$30/month
Same 100 GB, 80 GB accessed infrequently:
- 20 GB Standard: \$6/month
- 80 GB IA: \$2/month
Total: \$8/month (73% savings!)

- **EFS IA** = Infrequently accessed files (88% cheaper!)
- **Lifecycle policy** = Automatic transition (set to 30-90 days)
- **Not for all workloads** = If all files accessed daily, Standard is fine



Backup Strategies - Snapshots vs. Backups

EBS Snapshots:

Aspect	Details
What it is	Incremental backup of EBS volume
First snapshot	Full copy (all data)
Subsequent snapshots	Only changed blocks
Storage	Stored in S3 (but you don't see them)
Cost	Pay per GB-month of snapshot data

Snapshot Cost Optimization:

Tip	Savings
Delete old snapshots	No cost for what you don't keep
Snapshot lifecycle policies	Automatically delete after X days
Cross-region copy only for DR	Data transfer + storage costs
Use Amazon Data Lifecycle Manager (DLM)	Automate snapshot creation/deletion

AWS Backup vs. Native Snapshots:

Aspect	Native Snapshots	AWS Backup
Central management	✗ Per service	✓ Unified
Cross-service policies	✗ No	✓ Yes
Tag-based backup	✗ No	✓ Yes
Cost	Same (no additional fee)	Same (no additional fee)

AWS Backup (Centralized Backup Service):

Feature	Benefit
Central policies	One policy for EC2, RDS, DynamoDB, EFS, S3
Backup plans	Frequency, retention, cross-region copy
Tag-based backup	Backup all resources with specific tags
Cold storage tier	Move older backups to cheaper storage



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Data Transfer Optimization - Requester Pays and Transfer Acceleration

S3 Data Transfer Costs (memorize direction):

Transfer Direction	Cost
Upload to S3 (any region)	FREE
Download from S3 (same region)	FREE
Download from S3 (different region)	\$0.02-0.09/GB
Download from S3 (internet)	\$0.09-0.09/GB (varies by region)

Requester Pays Buckets:

Aspect	Details
What it does	Requester pays for data transfer and requests
Who pays	Requester (not bucket owner)
When to use	Sharing large public datasets
Example	Scientific data, research datasets



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Data Transfer Optimization - Requester Pays and Transfer Acceleration

S3 Transfer Acceleration:

Aspect	Details
What it does	Uses edge locations to speed up uploads
Best for	Cross-region uploads, poor internet
Cost	Additional \$0.04-0.08/GB (on top of standard)
When to use	Varies (test with S3 Transfer Acceleration Speed Comparison tool)

CloudFront Data Transfer Optimization:

Benefit	Savings
Caching	Reduces origin requests (S3)
Lower egress costs	CloudFront egress is cheaper than S3 egress
Free tier	1 TB per month free (first 12 months)

- **SAA Exam Focus:**

- Upload to S3 = FREE (always remember!)
- Requester Pays = For sharing large datasets
- CloudFront egress cheaper than S3 egress (use for public content)



Cost Optimization Tools - AWS Pricing Calculator, Cost Explorer

AWS Pricing Calculator:

Feature	Use
Estimate costs	Before deploying resources
Compare pricing models	On-Demand vs. Reserved vs. Savings Plans
Share estimates	URLs or PDFs

AWS Budgets:

Feature	Use
Custom budgets	Cost, usage, or reservation
Alerts	Email or SNS when exceeding threshold
Forecasted alerts	Alert if expected to exceed

AWS Cost Explorer:

Feature	Use
Analyze historical costs	12 months of data
Forecast future costs	Up to 12 months ahead
Filter by service, region, tag	Identify cost drivers
Rightsizing recommendations	Identify idle/underutilized resources

- **SAA Exam Focus:**

- **Pricing Calculator** = BEFORE deployment (estimate)
- **Cost Explorer** = AFTER deployment (analyze)
- **Budgets** = Alert when exceeding (proactive)



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Section 4.1 Summary - Cost-Optimized Storage

Service	Cost Optimization Strategy	Key Exam Point
S3 Standard	Use lifecycle to transition older data	Transition, not delete
S3 Standard-IA	30-day minimum duration	Early deletion fee
S3 Glacier Deep Archive	Cheapest, 180-day min, 12+ hour retrieval	Compliance, long-term
S3 Intelligent-Tiering	Automatic tiering for unknown patterns	Monitoring fee applies
S3 Lifecycle Policies	Automate transitions + expiration	Can't go Glacier → Standard
EBS gp3	Default choice (20% cheaper than gp2)	3,000 IOPS baseline
EBS Snapshots	Incremental (delete old snapshots)	First snapshot = full
EFS IA	88% cheaper for infrequent access	Lifecycle policy transitions
AWS Backup	Centralized backup management	Tag-based policies
S3 Transfer	Upload = FREE, download = cost	Requester Pays for sharing



Section 4.2: Cost-Optimized Compute

- **Topics Covered:**

- AWS Pricing Models (On-Demand, Savings Plans, Reserved Instances, Spot)
- Savings Plans Deep Dive (Compute vs. EC2 Instance Savings Plans)
- Spot Instances (up to 90% discount, interruptions, Spot Fleet)
- AWS Compute Optimizer (right-sizing recommendations)
- Right-Sizing Strategies (underutilized vs. over-provisioned)
- Serverless Cost Benefits (Lambda, Fargate - no idle cost)
- Hybrid Compute (Outposts, Snowball Edge)



AWS Pricing Models - The Big Picture

Four Pricing Models (MEMORIZE - heavily tested):

Model	Discount	Commitment	Flexibility	Best for
On-Demand	0%	None	Highest	Short-term, unpredictable, first-time
Savings Plans	Up to 72%	1-3 years	High (any instance, region, family)	Flexible, steady-state
Reserved Instances (RI)	Up to 75%	1-3 years	Low (specific instance)	Predictable, specific instances
Spot Instances	Up to 90%	None (can be interrupted)	High (but interruptions)	Fault-tolerant, batch jobs

- **SAA Exam Focus:**
 - **On-Demand** = Baseline (pay full price)
 - **Savings Plans** = Newer, more flexible than RIs
 - **RIs** = Older, less flexible, still valid
 - **Spot** = Cheapest, but can be taken away



Savings Plans Deep Dive - Compute vs. EC2 Instance Savings Plans

What are Savings Plans? Flexible pricing model offering lower prices in exchange for 1-3 year commitment to consistent compute usage (\$/hour)

Savings Plans Example:

Compute Savings Plan: \$10/hour commitment

- Can use: EC2 (any family/region) + Fargate + Lambda
- Total usage across all = \$10/hour

EC2 Instance Savings Plan: \$10/hour commitment for M5 in us-east-1

- Can use: M5 instances only in us-east-1
- Cannot use: C5, R5, or other regions

Two Types of Savings Plans (MEMORIZE differences):

Feature	Compute Savings Plans	EC2 Instance Savings Plans
Applies to	EC2, Fargate, Lambda	EC2 only
Instance family	Any (any family, any region)	Specific family (e.g., M5)
Region	Any region	Specific region
Instance size	Any (flexible)	Any size in that family
Discount	Up to 66%	Up to 72%
Flexibility	Highest	Limited (but higher discount)



Spot Instances - Up to 90% Discount

What are Spot Instances? Purchase unused EC2 capacity at steep discount (can be reclaimed)

Spot Instance Characteristics (MEMORIZE):

Aspect	Details
Discount	Up to 90% vs. On-Demand
Interruption notice	2-minute warning before reclaim
Pricing	Based on supply/demand (fluctuates)
Capacity	Not always available
Use case	Fault-tolerant, stateless, batch jobs

Spot Instance Interruption Handling:

Strategy	How it works
Don't use for critical	Never run production databases on Spot
Checkpointing	Save progress, resume later
Queue buffering	Store work in SQS before processing
Diversify	Use multiple instance types/families
Spot Fleet	Mix Spot + On-Demand for reliability



Spot Instances - Up to 90% Discount

What CAN run on Spot (Exam Favorites):

Workload	Why Suitable
Batch processing	Can restart from checkpoint
Data analysis (EMR)	EMR handles Spot interruptions
CI/CD workers	Transient workloads
Containerized apps	Fargate Spot even cheaper
Stateless web servers	Can spin up elsewhere

What CANNOT run on Spot:

Workload	Why Not Suitable
Production databases	Data loss risk
Stateful applications	Can't be interrupted
Real-time user requests	Interruption = bad UX
Long-running single instance	Risk of termination



AWS Compute Optimizer - Right-Sizing Recommendations

What is Compute Optimizer? ML-powered service that analyzes resource utilization and recommends optimal configurations

What Compute Optimizer Analyzes:

Service	Recommendations
EC2	Instance type (under/over-provisioned)
Lambda	Memory size
EBS	Volume type and size
Auto Scaling Groups	Launch template configuration

Compute Optimizer Data Sources:

Source	Details
CloudWatch metrics	CPU, memory, disk, network
4+ days of history	Minimum for recommendation
2 weeks recommended	Better recommendations
Cost Explorer integration	Shows cost impact

- **SAA Exam Focus:**
 - **Compute Optimizer** = Right-sizing (exam loves this)
 - **Minimum 4 days** = For recommendation (14 days ideal)
 - **Cost impact** = Shows expected savings



Right-Sizing Strategies - Underutilized vs. Over-Provisioned

Right-Sizing Process:

1. Monitor (CloudWatch) → 2. Analyze (Compute Optimizer) → 3. Rightsize → 4. Verify → 5. Repeat

Strategy	When to use
Scale down	Consistent low utilization
Change family	Wrong resource balance (memory vs. CPU)
Stop/terminate	Idle (dev/test over weekend)
Convert to Serverless	Intermittent workloads (Lambda/Fargate)
Use Spot	Fault-tolerant workloads
Use Savings Plans/RI	Steady-state, predictable

SAA Exam Focus:

- **Monitor first** (don't guess based on instance type)
- **Compute Optimizer** = Automated recommendation
- **Stop dev/test over weekend** = Easy savings (50-70% of week)



Serverless Cost Benefits - Lambda and Fargate

What makes serverless cost-effective? No idle resources (pay only when running)

Lambda Pricing Model:

Charged for	Rate	Free Tier
Requests	\$0.20 per 1M requests	1M requests/month
Compute (GB-seconds)	\$0.00001667 per GB-s	400,000 GB-s/month
Duration	Rounded to nearest ms	Included in compute

Fargate Pricing Model:

Charged for	Rate
vCPU per hour	\$0.04048 per vCPU-hour
Memory per GB-hour	\$0.004445 per GB-hour

SAA Exam Focus:

- **No idle cost** = Biggest serverless benefit (exam loves this)
- **Intermittent workloads** = Serverless wins
- **24/7 workloads** = EC2 with Savings Plans/RI wins



Hybrid Compute - AWS Outposts and Snowball Edge

What is Hybrid Compute? AWS services running on-premises (extend cloud to data center)

AWS Outposts:

Aspect	Details
What it is	Rack of AWS servers in your data center
How it works	AWS delivers, installs, maintains (you provide space/power)
Services	EC2, EBS, ECS, EKS, RDS, EMR
Pricing	Same as cloud (no data center markup)
Use case	Low-latency, data residency, on-prem apps

AWS Snowball Edge (Compute-Optimized):

Aspect	Details
What it is	Portable device with EC2 (for edge computing)
Use case	Remote locations (oil rigs, ships, military)
Compute	104 vCPUs, 416 GiB memory, optional GPU
Storage	80 TB HDD + 28 TB NVMe SSD

- **SAA Exam Focus:**

- **Outposts** = Permanent on-prem extension (keyword: "data residency", "low-latency")
- **Snowball Edge** = Portable edge computing (keyword: "remote", "oil rig", "ship")



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Section 4.2 Summary - Cost-Optimized Compute

Cost Optimization Principles:

Principle	Example
Use Spot when possible	Batch, EMR, CI/CD
Commit for steady workloads	Savings Plans (preferred) or RI
Right-size	Compute Optimizer
Stop idle resources	Dev/test over weekends (50-70% savings)
Prefer serverless for intermittent	Lambda, Fargate (no idle cost)
Monitor and adjust	Cost Explorer, Budgets



Section 4.3: Cost-Optimized Database Solutions

- **Topics Covered:**
 - Right-Sizing Databases (Compute Optimizer for RDS)
 - Aurora Serverless (pay per ACU-hour, intermittent workloads)
 - DynamoDB Capacity Modes (On-Demand vs. Provisioned)
 - When to Use Serverless Databases (development, unpredictable spikes)
 - Database Selection for Cost (DynamoDB vs. RDS vs. Aurora)
 - Data Retention Policies (archiving old data to S3)



Right-Sizing Databases - Compute Optimizer for RDS

- **What is RDS Right-Sizing?** Matching database instance size to actual workload requirements

Family	Use Case	Cost (db.r5.large vs. db.t3.large)
T (Burstable)	Dev/test, small production	Cheapest (but burst credits)
M (General Purpose)	Most production workloads	Moderate
R (Memory Optimized)	Memory-intensive (SAP, Oracle)	Higher
X (High Memory)	Large databases	Highest

RDS Compute Optimizer:

Feature	Benefit
ML-powered recommendations	Analyzes CloudWatch metrics
Cross-instance family comparisons	M5 vs. R5 vs. T3
Cost impact	Shows expected savings
Performance risk	Estimates impact of downsizing

SAA Exam Focus:

- **Right-size before buying RI** (don't commit to wrong size!)
- **Start small, scale up** (easier than over-provisioning)
- **CloudWatch metrics** = CPU, memory, IOPS (monitor for 2+ weeks)



Aurora Serverless - Pay Per ACU-Hour

What is Aurora Serverless? Aurora database that automatically scales compute capacity (pay only for what you use)

Aurora Serverless Pricing Model:

Charged for	Rate	How it works
ACU (Aurora Capacity Unit)	\$0.06-0.12 per ACU-hour	1 ACU = ~2 GB memory + corresponding CPU
Storage	\$0.10 per GB-month	Same as provisioned Aurora
I/O	\$0.20 per million requests	Same as provisioned Aurora

Aurora Serverless vs. Provisioned Aurora Cost Comparison:

Workload Pattern	Serverless	Provisioned	Winner
8 hours/day, weekdays	Pay for ~176 hours	Pay for 720 hours	Serverless (76% savings)
24/7 consistent	Pay for 720 hours	RI discount available	Provisioned (with RI)
Unpredictable spikes	Auto-scales	Over-provisioned	Serverless
Development (intermittent)	Scales to zero	Pay idle hours	Serverless



DynamoDB Capacity Modes - On-Demand vs. Provisioned

DynamoDB Capacity Modes:

On-Demand vs. Provisioned - When to choose each:

Scenario	Recommendation
Unknown traffic patterns	On-Demand (start, then evaluate)
Very low traffic (<10 requests/sec)	On-Demand (provisioned minimum expensive)
Predictable, steady traffic	Provisioned (30-50% cheaper)
Spiky but predictable peaks	Provisioned + Auto Scaling
Development/testing	On-Demand (pay only when used)

DynamoDB Pricing Units:

Unit	What it is	Cost (us-east-1)
RCU (Read Capacity Unit)	1 strongly consistent read/sec (4 KB)	\$0.00025 per RCU-hour
WCU (Write Capacity Unit)	1 write/sec (1 KB)	\$0.00065 per WCU-hour

• SAA Exam Focus:

- **On-Demand = Unpredictable** (keywords: "unknown", "variable", "new application")
- **Provisioned + Auto Scaling = Predictable** (keywords: "steady", "known pattern")
- **RCU/WCU = Know the definitions** (1 RCU = 4 KB/sec, 1 WCU = 1 KB/sec)



When to Use Serverless Databases - Decision Framework

Serverless Database Options on AWS:

Service	Serverless Option	Use Case
RDS	✗ No (only provisioned)	Traditional relational
Aurora	✓ Aurora Serverless v2	MySQL/ PostgreSQL, intermittent
DynamoDB	✓ On-Demand mode	NoSQL, unpredictable
Redshift	✗ No (RA3 provisioned)	Data warehouse

When to Choose Serverless:

Factor	Recommend Serverless	Recommend Provisioned
Traffic pattern	Unpredictable, intermittent	Predictable, steady-state
Workload	Dev/test, occasional batch	Production 24/7
Scaling needs	Unpredictable spikes	Predictable peaks
Cost sensitivity	Low traffic (<10% of time)	High traffic (24/7)
Cold start tolerance	Yes (few seconds)	No (needs immediate response)



Database Selection for Cost - DynamoDB vs. RDS vs. Aurora

Database	When to Choose (Cost Focus)	Why
DynamoDB	Simple lookups, massive scale, NoSQL	No idle cost (On-Demand), auto-scaling
DynamoDB Provisioned	Predictable NoSQL workload	Cheaper than On-Demand (30-50%)
RDS (MySQL/PostgreSQL)	Traditional relational, predictable	RI discounts available
Aurora	Relational, high performance, MySQL/PostgreSQL	5x performance, same price as RDS
Aurora Serverless	Intermittent relational workload	Pay per ACU-hour (no idle cost)
Redshift	Analytics, data warehouse	RA3 instances separate storage/compute

- **Key Insight (Exam Favorite):**
 - **Aurora costs the SAME as RDS** (provisioned instances)
 - **Aurora Serverless costs less** (intermittent workloads)
 - **DynamoDB = Pay per request** (not per instance)
- **SAA Exam Focus:**
 - **Aurora = Same price as RDS** (but 5x faster!)
 - **If both fit, choose Aurora** (better performance for same cost)
 - **RDS vs. DynamoDB = Relational vs. NoSQL** (not cost-driven)



Data Retention Policies - Archiving Old Data to S3

Why Archive Database Data? Move old, infrequently accessed data to cheaper storage

Archiving Options by Database:

Database	Archiving Method	Cost Savings
RDS/Aurora	Export to S3 (Parquet/CSV)	Remove from expensive storage
DynamoDB	Export to S3 (DynamoDB Export)	Remove RCU/WCU costs
Redshift	Unload to S3 (Parquet)	Reduce cluster size

Database Archiving Lifecycle:

Day 0-90: Database (fast query, expensive)
Day 91-365: S3 Standard (slower query via Athena)
Day 366+: S3 Glacier Deep Archive (12+ hour retrieval)

RDS Snapshot vs. Export to S3:

Feature	Snapshot	Export to S3
Queryable	✗ No (must restore first)	✓ Yes (Athena, Redshift Spectrum)
Granularity	Whole database	Per table/query
Cost for old data	Pay for snapshot storage	Pay for S3 storage (cheaper)
Best for	Disaster recovery	Data archiving + analytics

• SAA Exam Focus:

- **Export to S3 = Cheaper than keeping in database** (S3 costs less)
- **Athena = Query archived data** (serverless, no cluster)
- **Glacier Deep Archive = Long-term retention** (compliance, 12+ hour retrieval)



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Section 4.3 Summary - Cost-Optimized Database Solutions

Database Cost Optimization Checklist:

1	Choose right database (DynamoDB vs. RDS vs. Aurora vs. Redshift)
2	Right-size instance (don't over-provision)
3	Buy Reserved Instances for production (if 24/7)
4	Use Serverless for intermittent (dev/test, unpredictable)
5	Archive old data to S3 (Athena for query)
6	Monitor and adjust (Cost Explorer, Compute Optimizer)



Section 4.4: Cost-Optimized Network Architectures

- **Topics Covered:**
 - Understanding Data Transfer Costs (same AZ free, cross-AZ cost, cross-region higher, internet egress highest)
 - VPC Endpoints (Gateway Endpoint free for S3/DynamoDB vs. Interface Endpoint costs)
 - NAT Gateway vs. NAT Instance (managed vs. self-managed, cost comparison)
 - CloudFront Cost Benefits (caching reduces origin load, lower egress costs)
 - Direct Connect vs. VPN (consistent lower data transfer costs for large volume)



Understanding Data Transfer Costs - The Complete Picture

Transfer Path	Cost	When it applies
Same AZ (private IP)	FREE	EC2 to EC2, EC2 to RDS (same AZ)
Same region, different AZ	\$0.01/GB	Cross-AZ traffic (using private IP)
Same region, different AZ (public IP)	\$0.02/GB	If using public/internet IP
Different region	\$0.02/GB (plus region-specific)	Cross-region replication, DR
To internet (egress)	\$0.09-0.20/GB (first 1 GB free/month)	Serving users, API calls
From internet (ingress)	FREE	Uploads to S3, EC2, etc.

Key Insight for SAA:

- "Data transfer within same AZ is free"
- "Data transfer between AZs in same region costs"
- "Data transfer between regions costs"
- "Data transfer from AWS to internet costs"
- "Data transfer from internet to AWS is free"
- **SAA Exam Focus:**
 - **Ingress (upload) = FREE** (remember this!)
 - **Egress (download) = COST** (first 1 GB free)
 - **Keep resources in same AZ** → Free (when possible)
 - **Cross-AZ traffic costs** → Design to minimize



VPC Endpoints - Gateway Endpoint vs. Interface Endpoint Cost

VPC Endpoint Cost Comparison (CRITICAL for SAA)

Endpoint Type	Services	Cost	When to use
Gateway Endpoint	S3, DynamoDB only	FREE	Access S3/DynamoDB from VPC
Interface Endpoint (PrivateLink)	All other AWS services	\$0.01/hour + \$0.01/GB data	Access other services privately

Gateway Endpoint (FREE) - Use this when possible!

- Private Subnet EC2 → Gateway Endpoint (S3) → S3

(FREE, no NAT Gateway needed!)

When to use Interface Endpoint:

Scenario	Why Interface Endpoint
Access SQS, SNS, KMS, etc. privately	No NAT Gateway needed
Cross-account service access	PrivateLink connects accounts
On-premises to AWS services	Over Direct Connect

- SAA Exam Focus:**

- Gateway Endpoint = FREE** (always use for S3/DynamoDB!)
- Interface Endpoint = Cost** (\$0.01/hour + data)
- NAT Gateway = Most expensive** (avoid for S3 access)



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NAT Gateway vs. NAT Instance - Cost Optimization

NAT Gateway (Managed):

Aspect	Details
Hourly cost	\$0.045 per hour (~\$32.85/month)
Data processing cost	\$0.045 per GB
High availability	Built-in (within AZ)
Bandwidth	Up to 45 Gbps (auto-scales)
Management	None (AWS managed)

NAT Instance (Self-Managed):

Aspect	Details
Hourly cost	EC2 instance cost (e.g., t4g.nano ~\$0.0042/hour = ~\$3/month)
Data processing cost	Standard EC2 data transfer rates
High availability	You implement (scripting, Auto Scaling)
Bandwidth	Instance type dependent
Management	You manage (patching, scaling, failover)

- **SAA Exam Focus:**

- **NAT Gateway = Default choice** (managed, HA, cost-effective for most)
- **NAT Instance = Only for low volume + dev/test** (not production)
- **NAT Gateway data processing = \$0.045/GB** (cheaper than EC2 egress!)



CloudFront Cost Benefits - Caching and Lower Egress

Why CloudFront saves money:

Benefit	How it saves
Caching	Reduces requests to origin (S3, EC2, ALB)
Lower egress costs	CloudFront egress is cheaper than S3 egress
Data transfer to origin	Free (CloudFront to origin in same region)
Free tier	1 TB per month free (first 12 months)

CloudFront Cost Optimization Example:

Scenario: 10 TB/month served to users worldwide

Option 1: S3 direct

$10,240 \text{ GB} \times \$0.09 = \$921.60/\text{month}$

Option 2: CloudFront

$10,240 \text{ GB} \times \$0.085 = \$870.40/\text{month}$

Savings: \$51.20/month + caching benefits!

- **SAA Exam Focus:**
 - **CloudFront egress = Cheaper than S3 egress** (exam loves this)
 - **Caching = Reduced origin costs** (fewer S3/EC2 requests)
 - **Free tier = 1 TB/month** (great for small projects)



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Direct Connect vs. VPN - Cost for Large Volume Transfers

Cost Comparison for Large Data Transfer (10 TB/month to AWS):

Connection Type	Setup Cost	Monthly Cost (10 TB)	Best for
VPN (public internet)	\$0 (existing internet)	~\$921 (10 TB × \$0.09)	Low volume, quick setup
Direct Connect (1 Gbps)	\$0 (but requires carrier contract)	~\$200-400 (port + data)	High volume, consistent

- SAA Exam Focus:
 - VPN = Low volume, quick setup (minutes to hours)
 - Direct Connect = High volume, consistent (weeks to setup)
 - Ingress (to AWS) = FREE (both VPN and Direct Connect)
 - Egress (from AWS) = Cheaper with Direct Connect (for large volume)



Section 4.4 Summary - Cost-Optimized Network Architectures

Network Cost Optimization Checklist:

1. Keep traffic within same AZ when possible (free)
2. Use Gateway Endpoint for S3/DynamoDB (free!)
3. Use Gateway Endpoint for S3/DynamoDB (free!)
4. Choose NAT Gateway for most workloads
5. Consider Direct Connect for >5 TB/month egress
6. Monitor data transfer with Cost Explorer

The Golden Rules (MEMORIZE):

Rule	Why
Ingress is FREE	Uploads don't cost
Egress costs	Downloads cost
Same AZ is FREE	Keep things together
Gateway Endpoint is FREE	Use for S3/DynamoDB
CloudFront egress < S3 egress	Use CloudFront for public content



Final Exam Tips

- Top 10 Most Tested Concepts:

#	Concept	Why Important
1	IAM Roles vs. Users	Roles for services, users for people
2	Security Groups vs. NACLs	Stateful vs stateless, instance vs subnet
3	S3 Storage Classes	Know when to use each (Glacier for archive!)
4	RDS Multi-AZ vs. Read Replicas	HA vs. Performance
5	CloudFront vs. Global Accelerator	HTTP caching vs. TCP/UDP acceleration
6	Gateway Endpoint vs. Interface Endpoint	FREE vs. paid (S3/DynamoDB use Gateway!)
7	Savings Plans vs. Reserved Instances	Savings Plans more flexible
8	Spot Instances	Up to 90% discount, fault-tolerant only
9	SQS vs. SNS	Queue (one consumer) vs. Pub/Sub (many)
10	RPO vs. RTO	Data loss vs. downtime tolerance





GOOD JOB !!
TIME TO
TAKE A BREAK